

The ABBA_semiimplicit method

Implicit physical step and explicit application of its Jacobian

Adaptation to the GC2D problem with Hairer's symmetric projection

1 Purpose of the method

The `ABBA_semiimplicit` method advances two objects simultaneously:

$$z_n \in \mathbb{R}^2, \quad \xi_n \in \mathbb{R}^2,$$

where z_n is the physical state and ξ_n is an infinitesimal perturbation or tangent vector. The state is computed using the complete implicit ABBA method,

$$z_{n+1} = \Psi_{h,t_n}(z_n),$$

whereas the tangent vector is updated by applying the exact Jacobian of that same step:

$$\xi_{n+1} = D\Psi_{h,t_n}(z_n) \xi_n.$$

The term *semi-implicit* is used here in a precise sense:

- the trajectory $z_n \mapsto z_{n+1}$ requires solving for the implicit multiplier μ_n ;
- once μ_n has converged and the ABBA stages are known, the tangent dynamics is updated by means of a 2×2 linear system.

Essential distinction

The Jacobian formula does not allow the physical step to be replaced by

$$z_{n+1} = D\Psi_{h,t_n}(z_n) z_n.$$

This equality is false for a general nonlinear map. The correct application is

$$\delta z_{n+1} = D\Psi_{h,t_n}(z_n) \delta z_n.$$

Therefore, z_{n+1} is first computed with the implicit method, and its Jacobian is then applied to ξ_n .

2 GC2D system and the small matrix w

The physical state is $z = (X, Y)^T$ and satisfies

$$\dot{z} = f(z, t) = J_0 \nabla_z \phi(z, t), \quad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Let $s = h/2$. We introduce the *small* 2×2 matrix

$$w(z, t) := s D_z f(z, t) = s J_0 \nabla_z^2 \phi(z, t).$$

For the GC2D vector field this matrix is explicitly

$$w(z, t) = s \begin{pmatrix} -\phi_{XY}(z, t) & -\phi_{YY}(z, t) \\ \phi_{XX}(z, t) & \phi_{XY}(z, t) \end{pmatrix}. \quad (1)$$

Since the Hessian of ϕ is symmetric,

$$w(z, t)^T J_0 + J_0 w(z, t) = 0.$$

The factor s is included in w so that the Jacobian of each elementary ABBA stage has the particularly simple form “identity plus one w block.”

3 Implicit ABBA physical step

For a known state z_n and a provisional value $\mu \in \mathbb{R}^2$, the two copies are initialized as

$$u^{(0)} = z_n + \mu, \quad v^{(0)} = z_n - \mu.$$

The four A – B – B – A stages are then applied:

$$u^{(1)} = u^{(0)} + sf(v^{(0)}, t_n), \tag{2a}$$

$$v^{(1)} = v^{(0)} + sf(u^{(1)}, t_n), \tag{2b}$$

$$v^{(2)} = v^{(1)} + sf(u^{(1)}, t_{n+1}), \tag{2c}$$

$$u^{(2)} = u^{(1)} + sf(v^{(2)}, t_{n+1}). \tag{2d}$$

The multiplier for the step is the root $\mu_n = \mu_h(z_n)$ of the residual

$$\boxed{\mathcal{R}_n(z_n, \mu) := u^{(2)}(z_n, \mu) - v^{(2)}(z_n, \mu) + 2\mu = 0.} \tag{3}$$

Once the implicit iteration has converged, the next physical state is

$$\boxed{z_{n+1} = u^{(2)} + \mu_n = v^{(2)} - \mu_n.} \tag{4}$$

The two expressions in (4) coincide because $\mathcal{R}_n(z_n, \mu_n) = 0$.

4 Explicit calculation of the four w_i matrices

All quantities in this section are evaluated after the converged root μ_n has been obtained. The four small matrices are the matrix (1) evaluated at the four ABBA stage points:

$$w_1 := w(v^{(0)}, t_n) = s \begin{pmatrix} -\phi_{XY}(v^{(0)}, t_n) & -\phi_{YY}(v^{(0)}, t_n) \\ \phi_{XX}(v^{(0)}, t_n) & \phi_{XY}(v^{(0)}, t_n) \end{pmatrix}, \tag{5a}$$

$$w_2 := w(u^{(1)}, t_n) = s \begin{pmatrix} -\phi_{XY}(u^{(1)}, t_n) & -\phi_{YY}(u^{(1)}, t_n) \\ \phi_{XX}(u^{(1)}, t_n) & \phi_{XY}(u^{(1)}, t_n) \end{pmatrix}, \tag{5b}$$

$$w_3 := w(u^{(1)}, t_{n+1}) = s \begin{pmatrix} -\phi_{XY}(u^{(1)}, t_{n+1}) & -\phi_{YY}(u^{(1)}, t_{n+1}) \\ \phi_{XX}(u^{(1)}, t_{n+1}) & \phi_{XY}(u^{(1)}, t_{n+1}) \end{pmatrix}, \tag{5c}$$

$$w_4 := w(v^{(2)}, t_{n+1}) = s \begin{pmatrix} -\phi_{XY}(v^{(2)}, t_{n+1}) & -\phi_{YY}(v^{(2)}, t_{n+1}) \\ \phi_{XX}(v^{(2)}, t_{n+1}) & \phi_{XY}(v^{(2)}, t_{n+1}) \end{pmatrix}. \tag{5d}$$

For later use, define

$$\Sigma := w_2 + w_3. \tag{6}$$

5 Explicit calculation of the four stage Jacobians W_i

We now use a capital W_i for the 4×4 Jacobian of each elementary map in the duplicated variables (u, v) . Each one is an identity matrix plus exactly one small w_i block.

For the first A stage,

$$u^{(1)} = u^{(0)} + sf(v^{(0)}, t_n), \quad v^{(0)} = v^{(0)},$$

so

$$W_1 = \begin{pmatrix} I_2 & w_1 \\ 0 & I_2 \end{pmatrix} = I_4 + \begin{pmatrix} 0 & w_1 \\ 0 & 0 \end{pmatrix}. \quad (7a)$$

If $w_1 = (w_{1,ij})$, this means explicitly

$$W_1 = \begin{pmatrix} 1 & 0 & w_{1,11} & w_{1,12} \\ 0 & 1 & w_{1,21} & w_{1,22} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

For the first B stage,

$$v^{(1)} = v^{(0)} + sf(u^{(1)}, t_n), \quad u^{(1)} = u^{(1)},$$

therefore

$$W_2 = \begin{pmatrix} I_2 & 0 \\ w_2 & I_2 \end{pmatrix} = I_4 + \begin{pmatrix} 0 & 0 \\ w_2 & 0 \end{pmatrix}. \quad (7b)$$

Equivalently,

$$W_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ w_{2,11} & w_{2,12} & 1 & 0 \\ w_{2,21} & w_{2,22} & 0 & 1 \end{pmatrix}.$$

For the second B stage,

$$v^{(2)} = v^{(1)} + sf(u^{(1)}, t_{n+1}), \quad u^{(1)} = u^{(1)},$$

we obtain

$$W_3 = \begin{pmatrix} I_2 & 0 \\ w_3 & I_2 \end{pmatrix} = I_4 + \begin{pmatrix} 0 & 0 \\ w_3 & 0 \end{pmatrix}. \quad (7c)$$

Equivalently,

$$W_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ w_{3,11} & w_{3,12} & 1 & 0 \\ w_{3,21} & w_{3,22} & 0 & 1 \end{pmatrix}.$$

Finally, for the last A stage,

$$u^{(2)} = u^{(1)} + sf(v^{(2)}, t_{n+1}), \quad v^{(2)} = v^{(2)},$$

which gives

$$W_4 = \begin{pmatrix} I_2 & w_4 \\ 0 & I_2 \end{pmatrix} = I_4 + \begin{pmatrix} 0 & w_4 \\ 0 & 0 \end{pmatrix}. \quad (7d)$$

Equivalently,

$$W_4 = \begin{pmatrix} 1 & 0 & w_{4,11} & w_{4,12} \\ 0 & 1 & w_{4,21} & w_{4,22} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Thus the Jacobian of the complete four-stage ABBA map is the ordered product

$$\boxed{D\Phi_{h,t_n}^{\text{ABBA}} = W_4 W_3 W_2 W_1.} \quad (8)$$

The order cannot be changed because the matrices w_i , and therefore the matrices W_i , do not commute in general.

6 Construction of the physical Jacobian

Multiplying the four matrices in (8), the Jacobian on the duplicated phase space can be written in 2×2 block form as

$$D\Phi_{h,t_n}^{\text{ABBA}} = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

with

$$A = I_2 + w_4 \Sigma, \quad (9a)$$

$$B = w_1 + w_4 + w_4 \Sigma w_1, \quad (9b)$$

$$C = \Sigma, \quad (9c)$$

$$D = I_2 + \Sigma w_1. \quad (9d)$$

Notice that the factors s no longer appear separately in (9), because they are already contained inside every w_i .

To account for the implicit dependence $\mu_n = \mu_h(z_n)$, introduce the four 2×2 matrices

$$\begin{aligned} \mathcal{K} &= 4I_2 - (w_1 + w_2 + w_3 + w_4) \\ &\quad + (w_4 \Sigma + \Sigma w_1) - w_4 \Sigma w_1, \end{aligned} \quad (10)$$

$$\begin{aligned} \mathcal{L} &= w_1 - w_2 - w_3 + w_4 \\ &\quad + (w_4 \Sigma - \Sigma w_1) + w_4 \Sigma w_1, \end{aligned} \quad (11)$$

$$\mathcal{P} = I_2 + w_1 + w_4 + w_4 \Sigma + w_4 \Sigma w_1, \quad (12)$$

$$\mathcal{Q} = 2I_2 - w_1 - w_4 + w_4 \Sigma - w_4 \Sigma w_1. \quad (13)$$

Here,

$$\mathcal{K} = D_\mu \mathcal{R}_n, \quad \mathcal{L} = D_z \mathcal{R}_n.$$

Differentiating the identity

$$\mathcal{R}_n(z, \mu_h(z)) = 0$$

with respect to z gives

$$\mathcal{L} + \mathcal{K} D\mu_h(z) = 0,$$

and, provided that $\mathcal{K}$ is invertible,

$$D\mu_h(z_n) = -\mathcal{K}^{-1} \mathcal{L}.$$

Consequently, the exact Jacobian of the implicit physical step is

$$\boxed{J_n := D\Psi_{h,t_n}(z_n) = \mathcal{P} - \mathcal{Q} \mathcal{K}^{-1} \mathcal{L}.} \quad (14)$$

The matrix $\mathcal{K}$ in (10) is also the Jacobian of the residual used in Newton's method. However, $\mathcal{K}$ is not the physical Jacobian: the latter is J_n and also contains $\mathcal{P}$, $\mathcal{Q}$, and $\mathcal{L}$.

7 Applying the Jacobian without computing an inverse

In an implementation, $\mathcal{K}^{-1}$ should not be formed explicitly. To construct the entire Jacobian, solve the system with two right-hand sides:

$$\boxed{\mathcal{K}\mathcal{T} = \mathcal{L}, \quad J_n = \mathcal{P} - \mathcal{Q}\mathcal{T}.} \quad (15)$$

Since $\mathcal{K} \in \mathbb{R}^{2 \times 2}$, its factorization is inexpensive and can be reused for the two columns of $\mathcal{L}$.

If only a tangent vector ξ_n is to be propagated, it is not even necessary to form $\mathcal{T}$ or J_n . From (14),

$$J_n \xi_n = \mathcal{P} \xi_n - \mathcal{Q} \mathcal{K}^{-1}(\mathcal{L} \xi_n).$$

Therefore, compute successively

$$b_n = \mathcal{L} \xi_n, \quad \mathcal{K} y_n = b_n,$$

and finally

$$\boxed{\xi_{n+1} = \mathcal{P} \xi_n - \mathcal{Q} y_n.} \quad (16)$$

The only implicit part of (16) is a 2×2 linear system; no new nonlinear equation appears.

7.1 Explicit formula for the 2×2 system

Let

$$\mathcal{K} = \begin{pmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{pmatrix}, \quad b_n = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \quad \Delta_{\mathcal{K}} = k_{11}k_{22} - k_{12}k_{21}.$$

If $\Delta_{\mathcal{K}} \neq 0$, the solution of $\mathcal{K} y_n = b_n$ is

$$\boxed{y_n = \frac{1}{\Delta_{\mathcal{K}}} \begin{pmatrix} k_{22}b_1 - k_{12}b_2 \\ -k_{21}b_1 + k_{11}b_2 \end{pmatrix}.} \quad (17)$$

This expression makes it possible to apply (16) without a general-purpose matrix solver.

8 One-step algorithm

Given t_n , z_n , ξ_n , and h , one step of ABBA_semiimplicit is as follows:

1. Solve the nonlinear equation (3) to obtain μ_n . Each residual evaluation executes the four ABBA stages in (2).
2. With the converged root, recompute or retain the stages $u^{(0)}, v^{(0)}, u^{(1)}, v^{(1)}, v^{(2)}, u^{(2)}$.
3. Compute the new physical state using (4).
4. Evaluate explicitly w_1, w_2, w_3, w_4 using (5), and form $\Sigma = w_2 + w_3$.
5. If the full duplicated Jacobian is needed, construct the four stage matrices W_1, W_2, W_3, W_4 from (7) and multiply them in the order $D\Phi_{h,t_n}^{\text{ABBA}} = W_4 W_3 W_2 W_1$.
6. Construct $\mathcal{K}$, $\mathcal{L}$, $\mathcal{P}$, and $\mathcal{Q}$ using (10)–(13).
7. Compute $b_n = \mathcal{L} \xi_n$ and solve $\mathcal{K} y_n = b_n$.
8. Apply the physical Jacobian through $\xi_{n+1} = \mathcal{P} \xi_n - \mathcal{Q} y_n$.
9. Advance the time: $t_{n+1} = t_n + h$.

The output of the step is the pair

$$\boxed{(z_{n+1}, \xi_{n+1}).}$$

9 Propagation of the accumulated Jacobian

To obtain the derivative of the numerical solution with respect to the complete initial condition, propagate a matrix $M_n \in \mathbb{R}^{2 \times 2}$ instead of a single vector. Initialize

$$M_0 = I_2$$

and, after each physical step, update it according to

$$\boxed{M_{n+1} = J_n M_n.} \quad (18)$$

Thus,

$$M_N = J_{N-1} J_{N-2} \cdots J_1 J_0 = \frac{\partial z_N}{\partial z_0}.$$

The order of the products is important: the Jacobian of the most recent step always multiplies from the left.

The matrix update can also be performed without forming J_n . Solve

$$\mathcal{K}Y_n = \mathcal{L}M_n$$

for the two right-hand sides and set

$$\boxed{M_{n+1} = \mathcal{P}M_n - \mathcal{Q}Y_n.} \quad (19)$$

10 Symplecticity of the tangent propagation

If the projected ABBA step is solved exactly, its physical Jacobian satisfies

$$J_n^T J_0 J_n = J_0.$$

In two dimensions, this identity is equivalent to

$$\det J_n = 1.$$

Therefore, the application $\xi_{n+1} = J_n \xi_n$ is a symplectic linear transformation at each step. In addition, the accumulated product (18) is also symplectic:

$$M_N^T J_0 M_N = J_0, \quad \det M_N = 1.$$

In numerical computations, it is useful to monitor

$$\varepsilon_{\text{sp},n} := \left\| J_n^T J_0 J_n - J_0 \right\|, \quad \varepsilon_{\text{det},n} := |\det J_n - 1|.$$

These quantities will not be exactly zero because of roundoff, the tolerance used to solve (3), and errors in the derivatives of ϕ .

11 Interpretation in terms of two nearby trajectories

Let $z_n^\varepsilon = z_n + \varepsilon \xi_n$, with ε small. Differentiability of the implicit map gives

$$\Psi_{h,t_n}(z_n^\varepsilon) = \Psi_{h,t_n}(z_n) + \varepsilon D\Psi_{h,t_n}(z_n)\xi_n + \mathcal{O}(\varepsilon^2).$$

Since $z_{n+1} = \Psi_{h,t_n}(z_n)$ and $\xi_{n+1} = D\Psi_{h,t_n}(z_n)\xi_n$, it follows that

$$z_{n+1}^\varepsilon = z_{n+1} + \varepsilon \xi_{n+1} + \mathcal{O}(\varepsilon^2).$$

Thus, `ABBA_semiimplicit` simultaneously computes the reference trajectory and the linear evolution of an infinitesimal separation from it.

12 Implementation-ready summary

At each time step, perform the two updates

$$\begin{array}{ll} z_{n+1} = \Psi_{h,t_n}(z_n) & \text{using ABBA and the implicit root } \mu_n, \\ \xi_{n+1} = (\mathcal{P} - \mathcal{Q}\mathcal{K}^{-1}\mathcal{L})\xi_n & \text{using a } 2 \times 2 \text{ linear system.} \end{array}$$

The small matrices are

$$w_i = s D_z f(\text{stage point}_i, \text{stage time}_i),$$

and the four elementary 4×4 Jacobians are explicitly

$$W_1 = \begin{pmatrix} I_2 & w_1 \\ 0 & I_2 \end{pmatrix}, \quad W_2 = \begin{pmatrix} I_2 & 0 \\ w_2 & I_2 \end{pmatrix}, \quad W_3 = \begin{pmatrix} I_2 & 0 \\ w_3 & I_2 \end{pmatrix}, \quad W_4 = \begin{pmatrix} I_2 & w_4 \\ 0 & I_2 \end{pmatrix}.$$

Their ordered product is

$$D\Phi_{h,t_n}^{\text{ABBA}} = W_4 W_3 W_2 W_1.$$

The recommended tangent update remains

$$b_n = \mathcal{L}\xi_n, \quad \mathcal{K}y_n = b_n, \quad \xi_{n+1} = \mathcal{P}\xi_n - \mathcal{Q}y_n.$$

The physical state is not obtained by multiplying z_n by the Jacobian. The Jacobian is applied to the tangent vector or, equivalently, to the accumulated Jacobian with respect to the initial condition.